

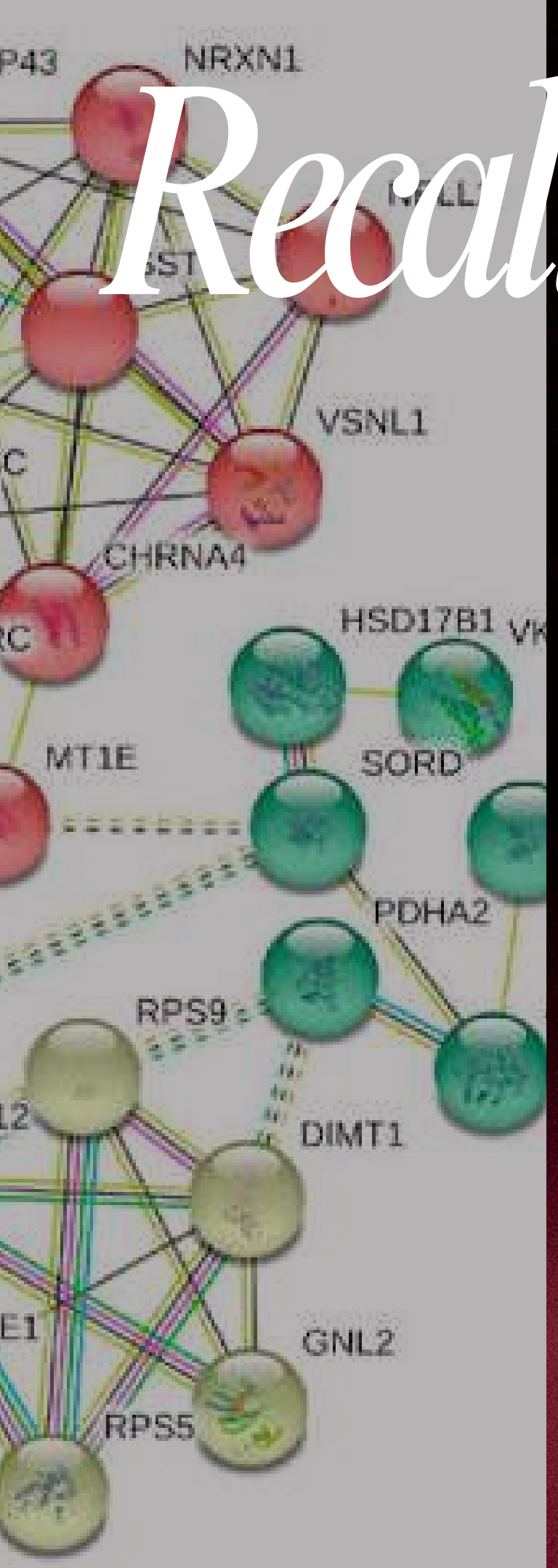
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BRIC research centre

Building the network

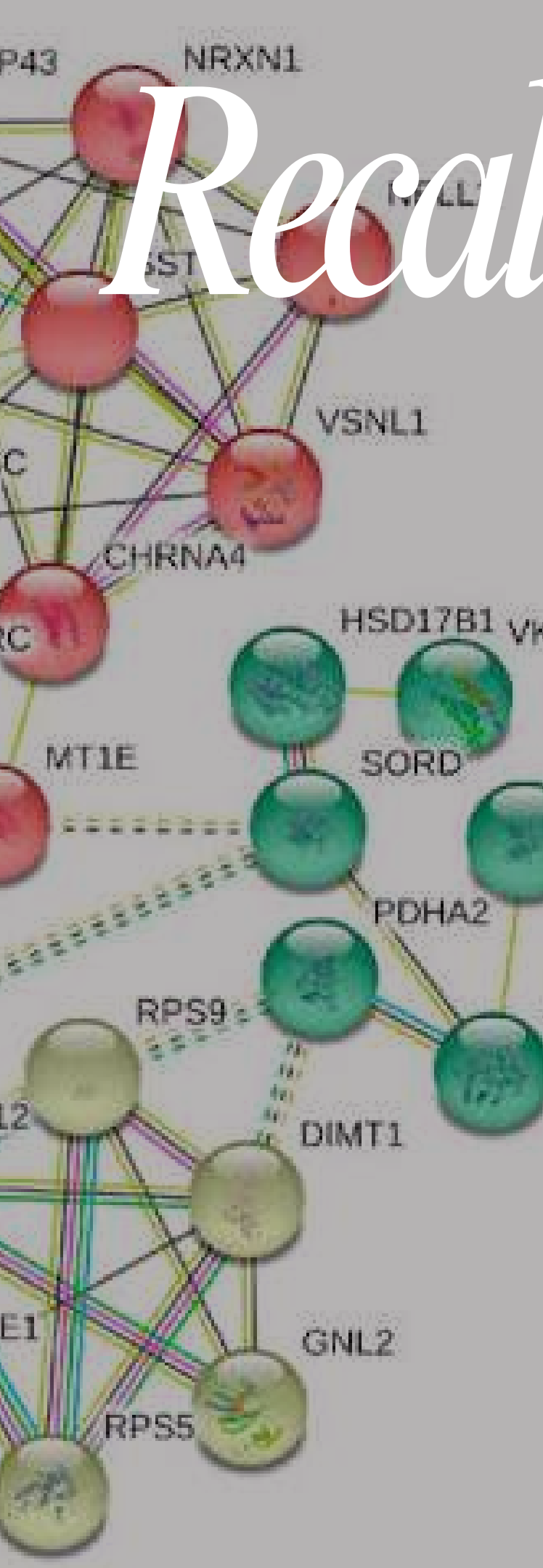
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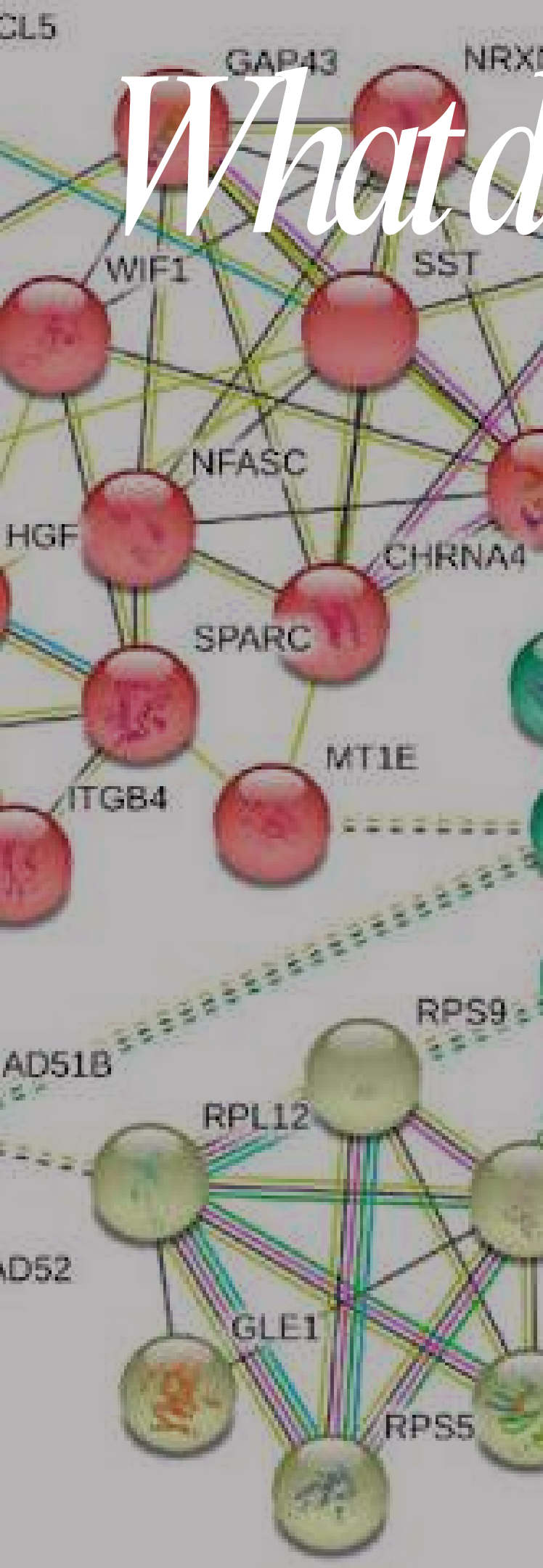
Recall: PPI networks

- **PPIs:** Physical and selective contacts that happen between pairs of proteins, in certain molecular regions and in a defined biological context
- **Interactome:** The totality of PPIs that happen in a specific biological context (e.g. cell, organism,..)
- **PPI network:** A graphical representation of a group of PPIs in which protein are represented as node and interactions as edges



Recall: Network properties

- **Scale-free:** The majority of nodes in a scale-free network have only a few connections to other nodes, whereas some nodes (hubs) are connected to many other nodes in a network. Biological networks are not random, nor are degrees equally distributed on the entire network.
- Scale-free networks are stable: If failures occur at random and most proteins have a low degree, the likelihood that a hub is affected is small.
- Many disease-linked proteins are hubs (e.g. P53)



What do we gain by analysing networks?

- Obtain information that cannot be obtained by analysing individual components
- Find key proteins such as hubs
- Find a key community of proteins (e.g. that belongs to the same pathway or complex)
- Identify entities involved in diseases
- Simulate *in silico* cell behaviour
- Formulate new hypotheses about protein function (guilt by association)

Why are pathways significant?

A biological pathway is an ordered series of molecular events that leads to the creation of a new molecular product, or a change in a cellular state or process. For example, a pathway could manufacture a protein, signal the presence of a hormonal signal, or cause a cell to move.

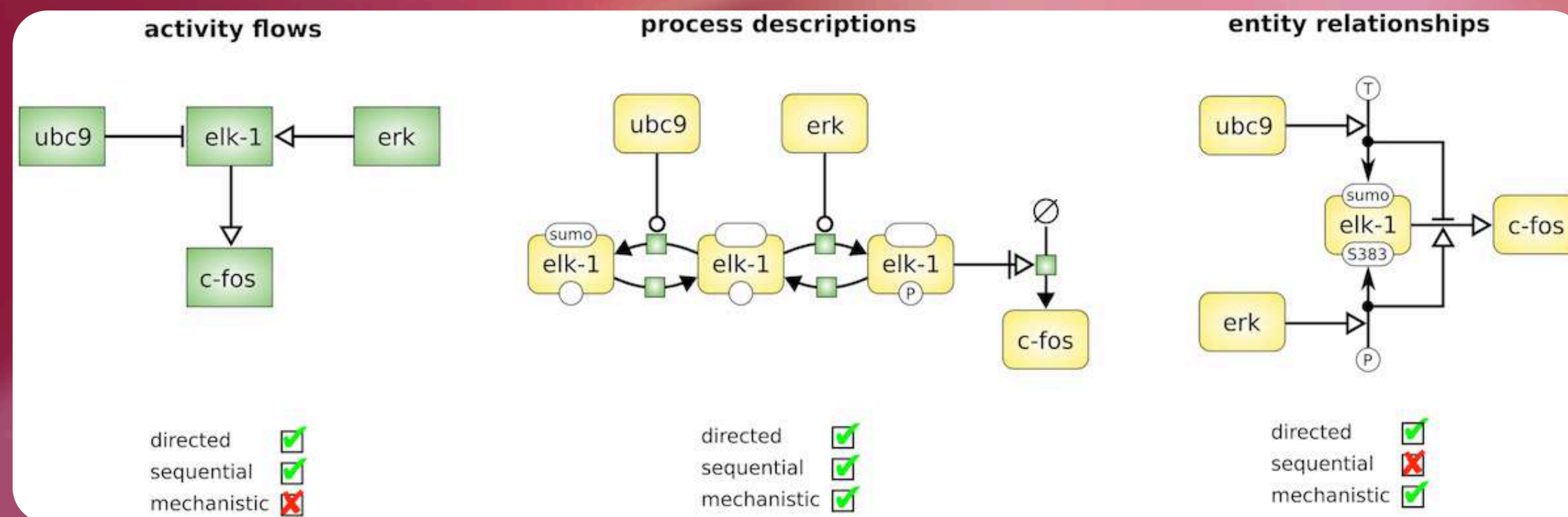
- A healthy cell (body) has many processes (pathways) that must all work together
- Faulty pathways can result in diseases such as cancer, asthma or diabetes. Most diseases are the result of small faults in several pathways, adding up to a problem that the body is unable to correct
- Medicines can correct faulty pathways, often by switching a signaling pathway on/off

Why do we need databases?

- Pathway information is often presented in scientific literature as text and diagrams. This triggers several limitations for scientists to reliably access and analyse this information because of the variety, volume and veracity in the data:
- The information is spread across many several hundreds of papers.
- Scientific exploration is often an ongoing work and information may be incomplete or outdated
- It is not easy to simultaneously investigate different types of biological information in pathways
- It is not accessible for computational reuse, i.e. the computer won't understand what to do with it.
- **"AKT1 physically interacts with GSK3 β to regulate downstream signaling." "TNF- α activates the NF- κ B signaling pathway." "PTEN negatively regulates the PI3K/AKT pathway."**

Systems biology graphical notation

- Every paper is basically telling us about one or a few interactions.
- Imagine having to comb through thousands of PPIs or gene interactions from hundreds of papers!
- The Systems Biology Graphical Notation (SBGN) project, an effort to standardise the graphical notation used in maps of biological processes.

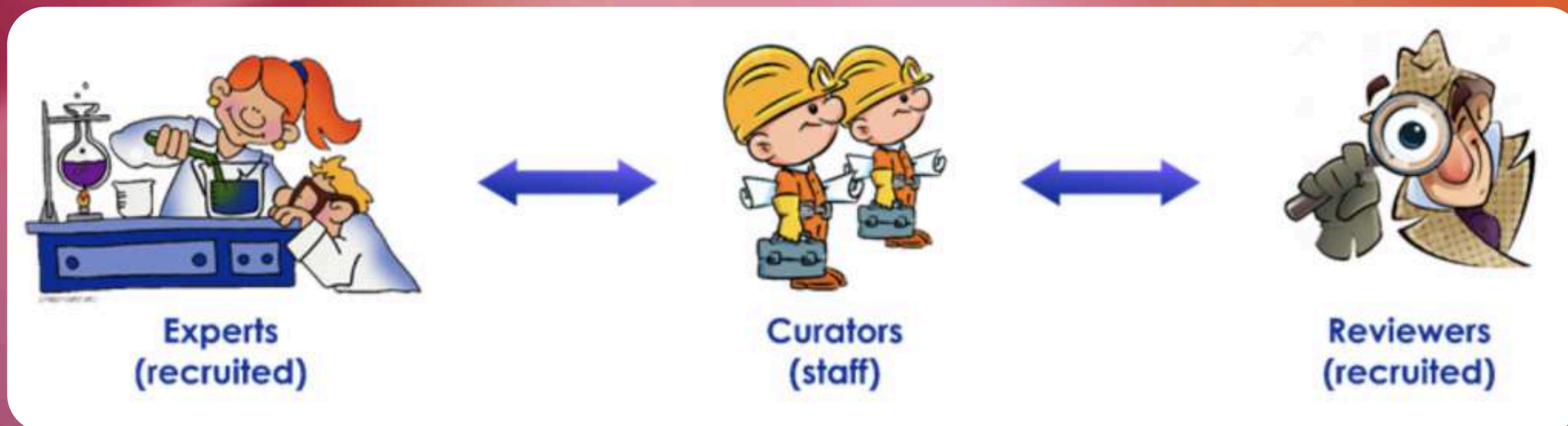


Database terminology

- **Reaction:** Any molecular event; including binding, phosphorylation, biochemical catalysis, molecular transport and spontaneous molecular events. These reactions can involve any biological molecule, but are typically between proteins and small molecules (chemical compounds).
- **Pathway:** A series of connected reactions, with each individual reaction considered as a single step in the pathway. These are represented in detailed molecular diagrams. Examples include immune system signalling, the cell cycle and apoptosis. Reactions are interconnected based on biological relevance resulting in the formation of pathways.
- **Curation:** Pathways are written by expert biologists and independently peer-reviewed before being included in the database. Reactions always cite scientific literature that experimentally verifies the described event. These manually curated human pathways are used as templates to computationally infer equivalent pathways in other species, including most commonly-used model organisms.

Database terminology

The starting point for a project is the recruitment of an expert in a particular area of biology. These expert authors are asked to describe pathways in their area and provide references. A curator works with the author to convert this information into the format used in the designated database. When the author and curator agree that the module is ready, it is sent for peer-review by another expert in the field. Following the review, the module is released publicly on the database. Databases are regularly updated.



Where does PPI data come from?

STRING

- most widely used resources for PPI networks.
- Includes both known and predicted interactions, plus functional enrichment tools.
- Useful for building network maps from gene/protein lists.

KEGG

- A major pathway and systems database that links molecular information to higher-level biological functions, including cellular processes, organisms, and ecosystems.

REACTOME

- A curated, peer-reviewed pathway database focused on reactions, pathways, and biological processes,
- Useful for pathway visualization and systems-level interpretation.

BIOGRID

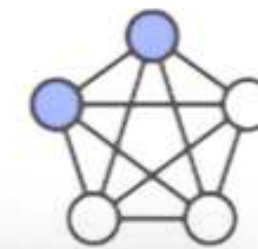
- Large repository of experimentally observed interactions.
- Covers PPIs, genetic interactions, and other functional associations.
- Good when you want curated interaction evidence from the literature.

INTACT

- Manually curated molecular interaction database.
- Strong for experimentally supported PPI data.
- Commonly used for high-confidence network construction and interactome analysis.

MINT

- Molecular Interaction database.
- Curates experimentally verified molecular interactions.
- Helpful for building reliable interaction networks.



STRING

STRING

- STRING scores every protein-protein interaction on a scale from 0 to 1. This score approximates the probability that a given interaction is a true biological interaction, given all the evidence STRING has for it. Higher score = more evidence = more confident (threshold = 0.7).
- Let's take a look!

Cytoscape

- A tool for network visualization and analysis.
- Contains a wide variety of advanced analysis and modelling tools, in the form of easily-installable plug-ins.

Interface

- **Visualisation panel/Canvas:** Shows the graphical visualisation of the network with a summary of nodes and edges
- **Network Panel:** This lists every network currently loaded in your session. If you load multiple networks, they all appear here as named entries. Click one to make it active. All operations (layout, styling, analysis) apply to whichever network is active. You can also create sub-networks: select a subset of nodes and go to File → New Network → From Selected Nodes. This creates an independent copy of just those nodes, which is useful when you want to work on a module separately.
- **Style Panel:** Controls the visual appearance of everything. It has three sub-tabs: Node, Edge, and Network. Each sub-tab is a table of visual properties — things like Fill Color, Size, Label, Border Width, Transparency.

Interface

- **Node Table:** This is a spreadsheet where every row is a node and every column is an attribute. When you import from STRING, you start with columns like name, description, and score. After you run Network Analyzer, new columns appear: Degree, BetweennessCentrality, ClusteringCoefficient, and more. You can sort by any column by clicking its header. You can also export this table to CSV for analysis. Clicking a row in the table selects that node on the canvas.
- **Edge Table:** Same as the Node Table but for edges. Contains the STRING combined score, the interaction type, and the individual evidence channel scores.
- **Filter Panel:** You can create filter rules to select nodes by attribute. For example: after Network Analyzer runs, you can filter to show only nodes where Degree is greater than 7.

01 | STRING

- Open STRING database
- Search for human protein TP53
- Export file as short tabular text output OR send network to cytoscape
- Import network on cytoscape
- Run network analyser and find hubs and bottlenecks
- Find first neighbour for TP53: Select → nodes → first neighbours of selected nodes

*Session
task 1*

Network analyzer results

- **Degree:** The simplest one, just counts the edges. TP53 with degree 9 means it has 9 direct interaction partners in this network. The degree you see in your network is only as large as your input gene list.
- **BetweennessCentrality:** For every pair of nodes in your network, there is a **shortest path** (the minimum number of hops to get from A to B). Betweenness centrality counts, for each node, how many of those shortest paths pass through it. A value of 0.61 for TP53 means that 61% of all shortest paths in the network pass through TP53. Information cannot flow from one part of the network to another without going through it. If you block TP53, you're severing the network.
- **ClosenessCentrality:** Measures how quickly a node can reach every other node. High closeness = short average distance to everyone. A node with closeness 0.9 can reach any other node in about 1.1 steps on average. This identifies proteins that can rapidly propagate signals throughout the entire network, useful for understanding which perturbations (like drug binding) will have the most systemic effects.

Network analyzer results

- **ClusteringCoefficient:** For each node, look at all its neighbours. How many of those neighbours are connected to each other? The clustering coefficient is that fraction. CDK2 with clustering coefficient 0.67 means that two-thirds of its neighbours are directly connected to each other. TP53 with a lower clustering coefficient despite high degree means its many neighbours don't connect to each other much, i.e. they're from different functional contexts that all converge on TP53 as a common regulator.
- **AverageShortestPathLength:** The mean number of hops from this node to every other reachable node. The inverse of closeness centrality, essentially. Useful for identifying nodes that are topographically central vs. peripheral.
- **NeighborhoodConnectivity:** The average degree of a node's neighbours. High value means this node connects to other well-connected proteins. Low value means it connects to peripheral, poorly-connected proteins.

Network analyzer results

- **Eccentricity:** The worst-case distance, the maximum shortest path from this node to any other node. Nodes with low eccentricity are near the centre of the network. This is used to calculate the network diameter (maximum eccentricity) and radius (minimum eccentricity), which are properties of the network as a whole.

02 | MERGING NETWORKS

- Import from NDEx:
- GO:0008585 (female gonad network)
- GO:0008584 (male gonad network)
- Tools → Merge → networks → intersection
- Tools → Merge → networks → differences
- Analyze common proteins

*Session
task 2*

1 | OBTAINING DATA

- Collect a list of genes involved in a particular disease (e.g. Rheumatoid arthritis)
- Import network into Cytoscape
- Run network analyzer
- Discuss top candidate hubs and bottlenecks

Deliverables:

- A screenshot of your network
- Your research question
- 1-2 paragraphs discussing your disease and results

*Project
task 1*

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Thank You!

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